

Interactive Data Fusion of Neural Radiance Fields for Industrial Facility Inspection in Virtual Reality

Ke Li ¹² *Susanne Schmidt ²Tim Rolff ²Reinhard Bacher ¹Wim Leemans ¹Frank Steinicke ²¹ Deutsches Elektronen-Synchrotron DESY, Germany² Human-Computer Interaction, Department of Informatics, Universität Hamburg

ABSTRACT

Real-world industrial facilities often present complex equipment that requires extensive inspection and maintenance for their operations. In this work, we present a virtual reality (VR) framework that supports virtual facility inspection and maintenance tasks by using neural radiance fields (NeRF) models to replicate, store, and visualize the appearance of complex industrial facilities. To overcome the performance bottleneck of VR NeRF rendering, we present two novel interactive data fusion techniques that can merge a NeRF model with its' corresponding CAD model through a mixed reality tunneling effect and contextual 3D NeRF drawing interaction. Technical benchmarking results and preliminary expert feedback are presented for the initial evaluation of our framework.

Index Terms: Virtual Reality—Human-computer Interaction—Neural Radiance Field—;

1 INTRODUCTION

The rapid development of virtual reality (VR) technology and the availability of affordable commercial VR head-mounted displays (HMDs) offer enormous potential for a wide range of industrial applications such as facility inspection and robot teleoperation. However, most existing industrial VR applications use conventional 3D data, such as Light Detection and Ranging (LiDAR) point cloud scans, textured meshes generated by photogrammetry, or computer-aided design (CAD) models to represent complex facilities. These 3D data provide limited functionalities for applications that require accurate replication of the real-world environment, as large-scale industrial facilities typically have surfaces and materials that are too complex to be modeled or reconstructed using photogrammetry or off-the-shelf RGBD sensors [5].

In recent years, neural radiance fields (NeRFs) [4] have emerged as a novel photorealistic 3D representation that introduces a paradigm shift of how the complex appearances of our realities can be replicated, stored, and visualized by a neural network learning only from 2D images and their camera poses [4]. However, rendering an image from a NeRF model requires an expensive ray tracing process that hinders its usage in a high-resolution, wide field of view (FoV) and high-framerate immersive VR applications. As a result, despite its enormous potential, applied research on VR-NeRF in real-world industrial settings remains sparse.

In this poster paper, we present an efficient VR NeRF framework to support virtual facility inspection for real-world industrial applications. Our framework presents two interactive data fusion and contextual visualization techniques to address the bottleneck of VR NeRF rendering. We evaluated our framework through technical benchmarking with real-world VR applications and expert

*e-mail: ke.li1@desy.de

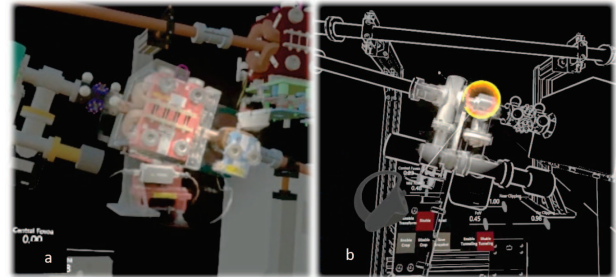


Figure 1: Screenshots from our VR facility inspection framework: (a) illustrates the data fusion of a high-resolution NeRF rendering merging with its' corresponding CAD model through mixed reality tunneling [2], and (b) illustrates the 3D NeRF drawing interaction using the CAD model as context.

feedback at large-scale particle accelerator facilities with the results confirming the effectiveness of our framework and demonstrating the empirical values of our data fusion methods for industrial VR applications.

2 METHOD

2.1 Data Fusion through FoV Restrictor

Applying a FoV restrictor at the peripheral region of the human eyes when users experience a VR application using a head-mounted display (HMD) introduces a “VR tunneling” effect. It is traditionally a popular technique for reducing users’ motion sickness during VR gaming but has also been used in mixed reality applications to perform data fusion of two different types of vision data for immersive HMDs [2]. As Figure 1 (a) shows, our first data fusion technique employs the design of such a FoV restrictor by displaying the NeRF model at the fovea to para-fovea region of the users’ eyes. To compensate for the undesirable effect that a FoV restrictor will likely decrease users’ immersion in the VR experiences and reduce their spatial orientation within the VR environments, the NeRF model is merged with its’ corresponding CAD model with the CAD model displayed at the peripheral region of the user’s eyes.

Our data fusion method with a FoV restrictor could balance the trade-off between high render quality and limited render performance of volumetric rendering. As the NeRF model does not need to be visualized at a wide FoV that matches the FoV of the HMD, it drastically decreases the number of rays and samples needed for VR NeRF rendering, therefore, reducing the total number of expensive queries to the neural network model during real-time VR application. Moreover, the desired FoV and render resolution can be flexibly adjusted according to the user’s preferences as well as the graphical device’s limitation, providing a saleable solution for visualizing large-scale real-world size NeRF models such as those captured at large-scale industrial facilities.

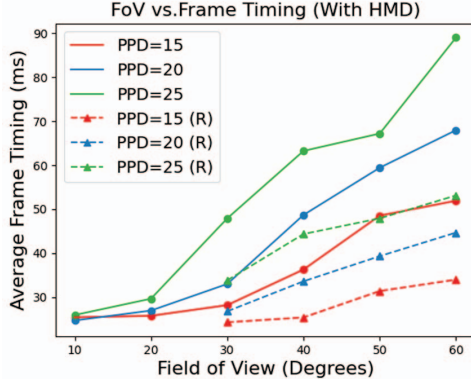


Figure 2: Systematic benchmark results showing the trend between rendering FoV and average frame timing for a test user following only approximately the same 3D trace. (R) indicates reduced rendering via 3D NeRF drawing.

2.2 Data Fusion through 3D NeRF Drawing

In addition to a FoV restrictor, our framework integrates a second data fusion design which enables a more robust and interactive selection of the render volume to be visualized via NeRF. As Figure 1 (b) illustrates, we design a 3D NeRF drawing effect, where users can edit the rendering region of the NeRF model by pointing a 3D sphere object that is attached to the virtual VR controller at a location where the NeRF model should be revealed or removed. The 3D NeRF drawing technique can improve the performance of VR NeRF rendering by taking advantage of the hierarchical sampling and empty space skipping optimization techniques that are integrated into typical NeRF rendering pipelines [3, 4]. The rendering volume that is “erased” by users is set to transparent by using a 3D bitfield mask and will be ignored by the ray tracer. As a result, the number of samples and queries to the neural network can be interactively adjusted and edited by users during application runtime. In a real-world application, such an interaction technique also enables industrial VR users to reveal the difference between their initial CAD designs and the actual implementations in reality.

3 PERFORMANCE BENCHMARK

We evaluate the performance of our system on a real-world industrial facility dataset consisting of 60 images reconstructing a section of a particle accelerator tunnel. Since the CAD model of the facility is designed to be at an exact scale of the real facility, we were able to estimate that the NeRF rendered volume of this model corresponds to approximately $2.2 \text{ m} \times 1.47 \text{ m} \times 2.08 \text{ m}$ real-world volume. We examine the performance trend when varying the NeRF rendering FoV from 10° to 60° . We also vary the pixel per degree (PPD) value to match the resolution requirements of VR displays of different quality, including medium display resolution (PPD=15), medium to high-end displays (PPD=20), or high-end displays (PPD=25). The final NeRF render resolution per eye ($R \times R$) for each configuration can be calculated as: $R = \text{FoV} \times \text{PPD} \times 2$, with the PPD multiplied by 2 to account for the upsampling required for aliasing reduction. The performance benchmark ran on an implementation based on the work of instant neural graphics primitives [4] and immersive neural graphics primitives [1]. All of our benchmarks ran on an Nvidia GeForce RTX 3090 GPU.

Figure 2 plots the average frame timing ($\bar{f}_t$) for each rendering configuration for both of our proposed data fusion methods. As expected, the average frame timing scales linearly as the FoV increases using the data fusion method through a FoV restrictor. Our

system can perform at 30 frames per second ($\bar{f}_t < 33.33\text{ms}$) at 30° FoV for devices with 20 PPD or at 40° at 15 PPD. Figure 2 also plots the average frame timing when rendering only selected areas defined via the 3D NeRF drawing technique. As expected, exposing only the selected render volume can reduce the overall frame timing. This confirms the performance optimization insight that removing unimportant scene details using the 3D NeRF drawing effect could be an option to gain additional performance for VR NeRF rendering.

4 EXPERT REVIEWS

Our framework and methods were evaluated through user studies and expert reviews at the Deutsches Elektronen-Synchrotron (DESY) with five facility control and management specialists. All participants confirmed that using NeRF for virtual facility inspection could benefit their workflow. One facility management expert commented that NeRF is a compelling, low-cost alternative for 3D facility documentation: *“I think the system has a good advantage. It is quite nice to project the NeRF model on the CAD model, as it is a lot more effort to take laser scans of the facility”*. Two experts mentioned that having a 1:1 scaled NeRF model aligned with the CAD model in immersive VR also gives them a better spatial awareness which could help them assess in advance if *“an operator’s hand would fit through a narrow gap to handle components”* or if special equipment would need to be prepared in advance. As commented by one participant who is leading the design and upgrade of particle accelerators: *“With this system, I see the possibility to test something in theory before you build it in practice. For example, when you have a machine, and you want to test if you have enough space to install it, it is quite nice you can test everything in the virtual area before you do it in reality.”*

5 CONCLUSION

In conclusion, this poster presents an effective VR framework for virtual facility inspection using NeRF. Through two novel interactive data fusion designs, our framework provides flexible and scalable VR NeRF rendering strategies that can balance the trade-off between high rendering quality and limited rendering performance. Technical benchmark results confirm the performance optimization capability of our methods. In addition, expert reviews at large-scale industrial facilities demonstrate the practical value of our framework in empirical settings. We encourage future research to extend our framework beyond virtual facility inspection to areas such as immersive version control, robot teleoperation, and remote telepresence systems.

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